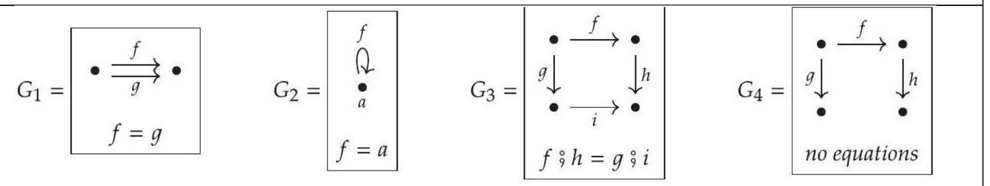
fong-spivak-invitation — formula report

4061 inline formulas (section omitted; 1 did not render), 262 display equations, 68 tables, 505 unrecovered image regions.

Unresolved (2)

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>fong-spivak-invitation_F02349</td><td>228</td><td>—</td><td>\operatorname{cospan} \underline{a} \xrightarrow{ \operatorname{id}_{a}} \underline{a} \stackrel{ \operatorname{id}_{\underline{a}}}{\text { 古 }}</td><td>(not rendered)</td><td>(iv) identities $\mathrm{id}_a \in \mathbf{Set}(a; a)$ just given by the identity cospan $\underline{a} \xrightarrow{\mathrm{id}_{\underline{a}}} \underline{a} \xleftarrow{\mathrm{id}_{\underline{a}}} \underline{a}$.</td></tr>
<tr><td>fong-spivak-invitation_EQ0202</td><td>291</td><td>—</td><td>G_{1}=\begin{array}{|c|} \hline \text { 李 } \\ f=g \end{array} \quad G_{2}=\begin{array}{|c|} \hline f \\ \stackrel{\mathcal{Q}}{a} \\ f=a \end{array} \quad G_{3}=\begin{array}{|ccc|} \bullet & \xrightarrow{f} & \bullet \\ g \downarrow & & \downarrow \\ \bullet & i \\ f \circ & \bullet \\ f \circ g \circ i \end{array} \quad G_{4}=\begin{array}{|cc|} \bullet & \stackrel{f}{ \longrightarrow} \bullet \\ g \downarrow & \\ \bullet & \bullet \\ \text { no equations } \\ \hline \end{array}</td><td>(not rendered)</td><td></td></tr>
<tr><td colspan="6">Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value
Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured</td></tr>
</table>

Flagged, not acted on — 7 of 76 shown

<table>
<tr><th>Identifier</th><th>Page</th><th>Conf.</th><th>LaTeX source</th><th>Rendered</th><th>Scan image</th></tr>
<tr><td>fong-spivak-invitation_ EQ0134 ● C|+91</td><td>213</td><td>■ 0.235</td><td><code>\begin{array}{rrrr} \operatorname{in}(\mu) & :=2, & \operatorname{in}(\eta):=0, & \operatorname{in}(\delta):=1, \end{array} \quad \operatorname{in}(\epsilon):=1,,</code></td><td>$\operatorname{in}(\mu) \;:=2, \quad \operatorname{in}(\eta):=0, \quad \operatorname{in}(\delta):=1, \quad \operatorname{in}(\epsilon):=1,,$</td><td>$\begin{array}{llll} \mathit{in}(\mu) := 2, & \mathit{in}(\eta) := 0, & \mathit{in}(\delta) := 1, & \mathit{in}(\epsilon) := 1, \\ \mathit{out}(\mu) := 1, & \mathit{out}(\eta) := 1, & \mathit{out}(\delta) := 2, & \mathit{out}(\epsilon) := 0. \end{array}$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0227 ● C|+43</td><td>306</td><td>■ 0.259</td><td><code>\begin{aligned} & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes\left(\mathrm{U}_{ \mathcal{X}} \times \eta_{\mathcal{X}}\right)((a, 1),(b, c, d)) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} \alpha^{-1}(x,(a, 1)) \otimes \mathrm{U}_{ \mathcal{X}}(a, b) \otimes \eta_{\mathcal{X}}(1, c, d)((b, c, d),(1, e)) \otimes \alpha((1, e), y) \\ = & \bigvee_{a, b, c, d, e \in \mathcal{X}} x(x, a) \otimes \epsilon_{\mathcal{X}}(b, c, 1) \otimes \mathrm{U}_{ \mathcal{X}}(d, e) \otimes \alpha((1, e), y) \\ = & x(x, y), \end{aligned}</code></td><td>$\begin{aligned} & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathcal{X}} \times \eta_{\mathcal{X}})\,((a,1),(b,c,d)) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathcal{X}}(a,b) \otimes \eta_{\mathcal{X}}(1,c,d)((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathcal{X}} x(x,a) \otimes \epsilon_{\mathcal{X}}(b,c,1) \otimes \mathrm{U}_{\mathcal{X}}(d,e) \otimes \alpha((1,e),y) \\ = & x(x,y), \end{aligned}$</td><td>$\begin{aligned} & \bigvee_{a,b,c,d,e\in\mathfrak{X}} \alpha^{-1}(x,(a,1)) \otimes (\mathrm{U}_{\mathfrak{X}} \times \eta_{\mathfrak{X}})((a,1),(b,c,d)) \\ & \qquad \otimes (\epsilon_{\mathfrak{X}} \times \mathrm{U}_{\mathfrak{X}})((b,c,d),(1,e)) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathfrak{X}} \alpha^{-1}(x,(a,1)) \otimes \mathrm{U}_{\mathfrak{X}}(a,b) \otimes \eta_{\mathfrak{X}}(1,c,d) \\ & \qquad \otimes \epsilon_{\mathfrak{X}}(b,c,1) \otimes \mathrm{U}_{\mathfrak{X}}(d,e) \otimes \alpha((1,e),y) \\ = & \bigvee_{a,b,c,d,e\in\mathfrak{X}} \mathfrak{X}(x,a) \otimes \mathfrak{X}(a,b) \otimes \mathfrak{X}(c,d) \otimes \mathfrak{X}(b,c) \otimes \mathfrak{X}(d,e) \otimes \mathfrak{X}(e,y) \\ = & \mathfrak{X}(x,y), \end{aligned}$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0075 ● C|+40</td><td>121</td><td>■ 0.380</td><td><code>\mathcal{G}:=\mid \stackrel{\text { Email }}{\underset{ \text { received_by }}{\stackrel{\text { sent_by }}{ \longrightarrow}}} \text { Address }</code></td><td>(not rendered)</td><td>$\mathcal{G} := \left| \begin{array}{l} \overset{\text{Email}}{\bullet} \underset{\text{received\_by}}{\overset{\text{sent\_by}}{\rightrightarrows}} \overset{\text{Address}}{\bullet} \\ \qquad \textit{no equations} \end{array} \right.$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0095 ● C|+37</td><td>146</td><td>■ 0.151</td><td><code>\operatorname{Col}(\Phi)(a, b):= \begin{cases} \mathcal{X}(a, b) & \text { if } a, b \in \mathcal{X} \\ \Phi(a, b) & \text { if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text { if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a, b) & \text { if } a, b \in \mathcal{Y}\end{cases}</code></td><td>$\operatorname{Col}(\Phi)(a,b) := \begin{cases} \mathcal{X}(a,b) & \text{if } a,b \in \mathcal{X} \\ \Phi(a,b) & \text{if } a \in \mathcal{X}, b \in \mathcal{Y} \\ \varnothing & \text{if } a \in \mathcal{Y}, b \in \mathfrak{X} \\ y(a,b) & \text{if } a,b \in \mathcal{Y} \end{cases}$</td><td>$\mathbf{Col}(\Phi)(a,b) := \begin{cases} \mathfrak{X}(a,b) & \text{if } a,b \in \mathfrak{X}, \\ \Phi(a,b) & \text{if } a \in \mathfrak{X}, b \in \mathfrak{Y}, \\ \varnothing & \text{if } a \in \mathfrak{Y}, b \in \mathfrak{X}, \\ \mathfrak{Y}(a,b) & \text{if } a,b \in \mathfrak{Y}. \end{cases}$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0131 ● C|+28</td><td>206</td><td>■ 0.985</td><td><code>\operatorname{colim}_{\mathcal{J}} D:=\{(v, d) \mid v \in V \text { and } d \in D(v)\} / \sim</code></td><td>$\operatorname{colim}_{\mathcal{J}} D := \{(v,d) \mid v \in V \text{ and } d \in D(v)\} / \sim$</td><td>$\operatorname*{colim}_{\mathfrak{J}} D := \big\{(v,d) \mid v \in V \text{ and } d \in D(v)\big\} / \sim$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0010 ● C|+21</td><td>044</td><td>■ 0.623</td><td><code>p \leq g(f(p)) \quad \text { and } \quad f(g(q)) \leq q .</code></td><td>$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$</td><td>$p \leq g(f(p)) \quad \text{and} \quad f(g(q)) \leq q.$</td></tr>
<tr><td>fong-spivak-invitation_ EQ0143 ● C|+20</td><td>226</td><td>■ 0.754</td><td><code>\begin{aligned} \circ_{i}: \mathcal{O} & \left(s_{1}, \ldots, s_{m} ; t_{i}\right) \times \mathcal{O} \left(t_{1}, \ldots, t_{n} ; t\right) \\ & \rightarrow \mathcal{O}\left(t_{1}, \ldots, t_{i-1}, s_{1}, \ldots, s_{m}, t_{i+1}, \ldots, t_{n} ; t\right) ; \end{aligned}</code></td><td>$\begin{aligned} \circ_i : \mathcal{O} & \left(s_1, \ldots, s_m; t_i\right) \times \mathcal{O}\left(t_1, \ldots, t_n; t\right) \\ & \rightarrow \mathcal{O}\left(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t\right); \end{aligned}$</td><td>$\begin{aligned} \circ_i \colon & \mathcal{O}(s_1, \ldots, s_m; t_i) \times \mathcal{O}(t_1, \ldots, t_n; t) \\ & \to \mathcal{O}(t_1, \ldots, t_{i-1}, s_1, \ldots, s_m, t_{i+1}, \ldots, t_n; t); \end{aligned}$</td></tr>
</table>

Conf. MathPix confidence: ■ ≥0.9 ■ 0.5–0.9 ■ <0.5 — no value

Residual render vs scan (inkdrill): • C component • W weak • S stable • N noise • K clean • not measured

The remaining 69 flagged rows are stated as a count.

Each differs from its scan by fewer than 20 components, which is the tail of the distribution rather than its bulk: 40 are component (`C`), 29 weak (`W`). By MathPix confidence: 37 at $\geq$0.9, 12 in 0.6–0.9, 20 below 0.6, 0 with none. Out/465 sampled twenty rows drawn from the $\geq$0.9 band corpus-wide and found eighteen purely typographic differences, one typographic plus a crop overrun, one borderline and *no* unambiguous content errors — so a high-confidence flag here is evidence of a rendering difference, not of a defect. Every row is in `report.ink.json`.